

DevOps Day-6

JAVA APPLICATION DEPLOYMENT IN MINIKUBE

```
pipeline {
    agent any

    stages {
        stage('SCM') {
            steps {
                git branch: 'main', url: 'https://github.com/Pavish-
kumar/guvidevopswarsday1'
            }
        }
        stage('Build-clean') {
            steps{
                sh 'mvn clean'
            }
        }
        stage('Build-validate') {
            steps{
                sh 'mvn validate'
            }
        }
        stage('Build-compile') {
            steps{
                sh 'mvn compile'
            }
        }
    }
}
```

```

stage('Build-test') {
    steps{
        sh 'mvn test'
    }
}
stage('Build-package') {
    steps{
        sh 'mvn package'
    }
}
stage('build to images') {
    steps {
        script{
            sh "docker build -t pavishkumar/webapplication ."
        }
    }
}
stage('docker push hub') {
    steps {
        script{
            withDockerRegistry(credentialsId: 'cred-1', url:
'https://index.docker.io/v1/') {
                sh 'docker push pavishkumar/webapplication'
            }
        }
    }
}
stage('Deploy App'
{ steps

```

```
{
  withKubeConfig(caCertificate: "", clusterName: 'minikube', contextName:
'minikube', credentialsId: 'cred-2', namespace: "", restrictKubeConfigAccess:
false, serverUrl: 'https://192.168.39.226:8443') { sh 'kubectl apply -f
Deployment.yml --validate=false' }
}
}
}
}
```

Deployment.yml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-deploy
  labels:
    name: my-deploy
spec:
  replicas: 1
  selector:
    matchLabels:
      apptype: web-backend
  strategy:
    type: RollingUpdate
  template:
```

```
metadata:
  labels:
    apptype: web-backend
spec:
  containers:
  - name: my-app
    image: praneshc/webapplication
    ports:
    - containerPort: 9000
```

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
  labels:
    app: my-service
spec:
  type: NodePort
  ports:
  - port: 9000
    targetPort: 8080
    nodePort: 30002
  selector:
    apptype: web-backend
```

```
pavis@pavish-kumar: ~  
service list'  
  
pavis@pavish-kumar:~$ ls  
docker-compose.yml  guvidevopswarsday1  my-pod.yaml  
pavis@pavish-kumar:~$ pwd  
/home/pavis  
pavis@pavish-kumar:~$ sudo nano my-pod.yaml  
[sudo] password for pavis:  
pavis@pavish-kumar:~$ sudo nano Deployment.yml  
pavis@pavish-kumar:~$ kubectl apply -f Deployment.yml  
deployment.apps/my-deploy created  
service/my-service created  
pavis@pavish-kumar:~$ kubectl get pod  
NAME                READY   STATUS    RESTARTS   AGE  
my-app              1/1     Running   4 (14m ago)  2d  
my-deploy-64b6d4dd7f-65cf5  1/1     Running   0           9s  
my-pod              1/1     Running   4 (14m ago)  2d  
pavis@pavish-kumar:~$ minikube service my-service  


| NAMESPACE | NAME       | TARGET PORT | URL                       |
|-----------|------------|-------------|---------------------------|
| default   | my-service | 9000        | http://192.168.58.2:30002 |

  
🔗 Starting tunnel for service my-service.  


| NAMESPACE | NAME       | TARGET PORT | URL                    |
|-----------|------------|-------------|------------------------|
| default   | my-service |             | http://127.0.0.1:33343 |

  
🖱 Opening service default/my-service in default browser...  
🖱 http://127.0.0.1:33343  
! Because you are using a Docker driver on linux, the terminal needs to be open to run it.
```

javaapplication [Jenkins]

Pipeline Syntax: Snippet

New cloud [Jenkins]

docker hub - Search

pavishkumar (Pavish Kumar)

localhost:8080/job/javaapplication/

Gmail YouTube GitHub Dashboard Cisco Networking A... TCS ION: Transform... Zoho Corp Careers

Jenkins

log out

Dashboard > javaapplication >

Status

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

Builds

Filter

🟢 javaapplication

Add description

Stage View

	SCM	Build-clean	Build-validate	Build-compile	Build-test	Build-package	build to images	docker push hub
Average stage times: (full run time: ~48s)	2s	3s	2s	3s	3s	4s	1s	6s
#3 13:40 No Changes	1s	942ms	1s	2s	2s	3s	1s	31s
#4 13:38 No Changes	1s	2s	2s	2s	2s	3s	1s	513ms failed
#2 13:37 No Changes	1s	1s	2s	2s	2s	4s	985ms	471ms

